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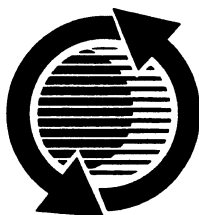
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ABSTRACT

An integrated lightweight structure and thermal insulation has been developed for the Mars Rover Warm Electronics Box (WEB). It combines a low thermal conductivity fiber reinforced composite structure with co-cured thermal control surfaces and an ultralightweight hydrophobic solid silica aerogel which eliminates convection and minimizes conduction and radiation. It provides excellent thermal insulation both at low gas pressures and in vacuum; and meets the structural requirements for spacecraft launch loads and for a 60 g impact landing at Mars without damage to the insulation or structure. The integrated design provides for variable conductance to meet the different thermal requirements for the different phases of operation. The system incorporates heat conduction paths to provide thermal control on the ground and during interplanetary cruise, which are then disconnected for Mars operation. The thermal performance of the design was evaluated at prototype level, and as a full structural engineering model and flight system. The thermal conductivity for the structure with insulation was measured at the prototype level to have a value of 0.033 W/mK and 24 °C at 1 atmosphere pressure, and in a 10 torr CO₂ simulated Mars environment to be 0.0163 W/mK at 24 °C, 0.0126 W/mK at -27 °C and 0.0089 W/mK at -80 °C.

INTRODUCTION

The Microrover Flight Experiment (MFEX) is part of the scientific and engineering payload for the Mars Pathfinder mission to explore the Martian surface. Pathfinder is the second Discovery mission, a NASA initiative for low cost focused planetary exploration. The Pathfinder mission is intended to provide an engineering demonstration of key technologies and concepts for the future exploration of Mars employing surface landers and rovers. The mission is scheduled for launch in December 1996 on a Delta II and is expected to land on Mars on July 4, 1997. Upon arrival at Mars, the

Pathfinder spacecraft will enter the Martian atmosphere, and then deploy in sequence a parachute, retrorockets, and then airbags to dissipate orbital energy for a controlled landing on the Mars surface. The Mars Rover will then be deployed for its primary mission to explore the Mars surface. Its primary exploration phase is scheduled to be 7 Martian days, with a design life of 30 days. A drawing of the full rover system is shown in Figure 1.

The Mars rover is limited to a total mobile mass of 11.5 kg, including the Alpha-Proton X-ray Spectrometer (APXS) scientific instrument, ten engineering experiment packages and an imaging system. The basic WEB structure is contained within a volume of 340 mm by 275 mm by 150 mm. The deployed rover size is 650 mm by 470 mm by 320 mm. It will have autonomous navigation, using a pair of wide angle imaging cameras for obstacle detection and avoidance. Its mobility will be provided by a six wheel drive, rocker-bogie suspension system. It will maintain two-way contact with the Pathfinder lander via an ultra-high frequency (UHF) link with the lander. Power will be supplied from a single 0.20 m² solar panel that will provide a peak power of 16 Watts and a non-rechargeable battery system that can provide 150 W-hr at 50 % depth of discharge. The Mars Rover thermal subsystem consists of a Warm Electronics Box (WEB), three RHU heater units to provide a constant 2.9 Watts heat input, and various heater elements on the batteries, modem and actuators. The WEB must vent the air pressure during launch, survive launch and landing, and meet strict planetary protection standards. The WEB also has electrical cabling passing through the cable tunnel to connect with the solar array, motors, charge couple device (CCD) cameras, the drive system and other external experiments. The Mars Rover design developed from the need to meet conflicting thermal requirements, the need to stay within system mass allocation without giving up functionality and the requirement to stay within a small volume envelope.

THERMAL REQUIREMENTS AND ENVIRONMENTS

The overall thermal system requirements are based on the internal temperature requirements of the electronic components during the different phases of operation environments. The Mars Rover utilizes standard industrial and military electronic components. Based on military specifications, the electronics are designed to operate in the range between $\pm 40^{\circ}\text{C}$. These limits are additionally constrained by the battery temperature requirements. The batteries are to be maintained at 40°C or less for the 3 months at the launch pad, 30°C for 8 months during cruise, and -30°C to 55°C (with less than 5 hours at above 40°C) on the Mars surface.

A significant constraint that challenged the design are the conflicting thermal environments that occur during the phases of ground handling prior to launch, during cruise when the Rover is encapsulated within the Pathfinder lander, and the first day on Mars when the Rover is still attached to the Pathfinder structure, and its primary mobile mission. The concern for ground handling operations is that the Rover batteries might become too warm in the interim prior to launch after the RHU's have been installed. This will be alleviated by special air conditioning during ground operations. During cruise, the Rover may become too hot due to the warm environment within the Pathfinder lander structure. Heat will be removed by a thermal link from a rover structural attachment to the lander petal. The Pathfinder lander will have an active cooling system to remove waste heat during cruise. This is the primary thermal link of the Rover with the Lander. There is no power or other direct electrical link with the Rover to the Lander. This thermal link poses a potential problem for the first day on Mars. The Rover is expected to still be attached in its stowed position to the Lander for up to 24 hours after landing on Mars. In this configuration, the thermal link is still in place, and the lander petal acts as a large thermal heat sink prior to the Rover being released for its operational phase being mobile on the Martian Surface.

There are 5 distinct thermal environments in which the Rover must maintain the required internal temperature range. They are:

1. Earth operations (no RHU's)
2. On the launch pad (with RHU's)
3. Cruise (Radiation dominated vacuum, RHU's, and cold finger)
4. Attached to the lander on Mars (8 Torr CO_2 , -81°C to -23°C , 0 to 12 m/s wind, and cold finger)
5. Mars exploration (8 Torr CO_2 , -81°C to -23°C , 0 to 12 m/s wind)

These five phases of operation drove the design and characteristics of the Mars Rover. Different mechanisms of heat transfer dominate during each of these five

distinct thermal environments. On the ground, convection and conduction are the primary mechanisms to maintain the system within the temperature limits. During cruise, there is only conduction through the structure, radiative coupling to the interior of the lander, and the conductive path through a thermal link to a cold finger that is part of the Pathfinder cooling fluid loop. On Mars at 8 torr atmospheric pressure, the wind from the CO_2 environment is still in the free molecular flow range and produces convective losses.

The Mars Rover mission operation scenario is driven by the Martian environment and the power limitations on the Rover for heat generation. The power management philosophy is to use the solar array power to operate the Rover and to use the excess power to heat up the internal components inside the WEB. No battery power will be used to maintain internal temperatures as it is reserved for emergency surge and nighttime operations. Throughout daily operations excess solar power will be dumped, via film heaters, to the batteries, which are the most significant thermal mass in the WEB. Because of the limited power available, parasitic losses place severe constraints on the insulation to meet the required temperature ranges.

INSULATION HISTORY - The last United States program to place a lander on Mars was the Viking program in the early 1970's. The thermal insulation concepts evaluated for the Viking program were fiberglass insulation, organic foams, colloidal powders, and multilayer insulation blankets[1]. These insulation systems were evaluated for thermal conductivity, heat sterilization, launch venting, vibration and shock integrity and Mars surface environmental stability. The Viking program eventually used a mixture of organic foam and fiberglass insulation, and multilayer insulation.

The initial insulation's investigated in the present work were:

1. Owens-Corning Aura™ vacuum jacketed fiber insulation
2. Polyurethane Foam
3. Opacified Aerogel powder encapsulated in Nomex Honeycomb core

The Owens-Corning Aura™ is essentially a nonrigidized vacuum bottle concept of thin metal sheets sealed around evacuated fiberglass insulation[2]. Although this design had low conductivity, the edge effects of individual small panels were impractical and additional mass would have been needed for a structure to give it the necessary rigidity. The polyurethane foam concept, being incorporated into the Pathfinder lander, was too bulky to be used for the Mars Rover because of strict volume and mass constraints. Furthermore, both of these types of insulation required separate structural penetrations to support payloads. These penetrations would be significant heat leaks. From previous

studies[1] it was known that colloidal silica had one third lower thermal conductivity than conventional foam insulation. Since the 1970's, particulate silica aerogel powder's have become available on the market which have better insulating properties than colloidal powders. An initial concept for integrated structure and thermal control was to build a honeycomb sandwich structure for the WEB with the honeycomb volume filled with aerogel powder. A short developmental effort was undertaken using Syloid 244 aerogel powder from the W.R. Grace Company. An attempt was made to improve the thermal performance of this concept by the addition of opacifying particulate to the aerogel powder to reduce internal radiation transport. The opacifying materials evaluated ranged from carbon blacks to aluminum powders. An optimized thermal system was developed incorporated micron size aluminum powder to opacify the silica aerogel. This system had a thermal conductivity of 0.018 W/mK at 25°C in a 10 torr CO₂ environment. Since silica aerogel powders were not available in extremely low densities and the addition of the opacifying agent increased the overall bulk density to 160 mg/cc, the resulting insulation weight exceeded the mass allocation. A new approach had to be taken. Solid aerogel was the best way to reduce the insulation mass significantly.

In the last decade, there has been a significant amount of interest in solid aerogel materials, both organic and inorganic[3]. This is a result of their low densities, high surface areas and other unique properties. One unique property is the low thermal conductivity of solid aerogels. Fricke[4] has studied the thermal properties of aerogels, and has shown that the thermal conductivity of silica aerogel is dependent both on its density and environment. He was able to express the experimental total thermal conductivity of silica aerogel in terms of its conductive, convective and radiative components by analysis. This experimental and analytical result is illustrated in Figure 2 for silica aerogel at 25°C and 1 atmosphere air pressure. Fricke did not study aerogel properties down to the very low densities that this program is using. The Mars Rover operates both at reduced pressures and for lower temperatures. For both lower temperatures and pressures the effective thermal conductivity for silica aerogel was expected to be lower than at standard temperature and pressure conditions. The solid silica aerogel also showed less sensitivity for thermal conductivity to pressure variations from 8 torr to vacuum than the particulate silica aerogel. Based on these considerations and the significant mass reductions that that could be obtained, solid silica aerogel was incorporated into the thermal design.

INTEGRATED STRUCTURE AND THERMAL DESIGN

One aspect of the small size and volume of the Mars Rover design is that interior components such as

circuit boards and the structural axial tube which spans from side wall to side wall. The structural axial provides the mechanical attachment point for the rocker-bogie wheel assembly and transmits most launch and landing structural loads, and provides the housing for the RHU's. It is designed out of glass-epoxy to meet structural requirements and minimize conductive losses. Attachments to the walls are through thermally isolating composite fittings. The design feature of directly attaching components to the walls was utilized to integrate the interior components with the structural design. While this feature has the advantage of reducing the mass required for the structure, it also has the unfortunate effect of providing a direct thermal heat loss path for the electronic components. This led the Rover team to develop an integrated structure and thermal design for the Mars Rover which minimizes these heat losses. This design can best be described as a sheet and spar design for the WEB, similar to that used in aircraft design. It incorporates thermally insulating solid aerogel within a lightweight composite structure.

The sheet and spar mechanical design is composed of a set of E-glass/epoxy structural members, with each volume filled with low density solid silica aerogel. The basic construction is illustrated in Figure 3. E-glass/epoxy was chosen as the structural material based on a trade study that evaluated the specific strength, stiffness, and thermal conductivity of composite materials. The other materials evaluated were S-glass/epoxy, Kevlar 49 /epoxy and Spectra/epoxy composite materials. The E-glass/epoxy system was chosen for its mechanical and thermal properties, as well as its low cost and availability. The Kevlar system had excellent properties but would have been difficult to integrate with the aerogel because of its high stiffness. The Spectra system would not have been able to meet the thermal sterilization requirements of 125°C for 5 hours, necessary to meet planetary biological protection requirements. Figure 4 presents an isometric view of the Rover WEB. The thickness and therefore stiffness of the facesheets and spars were varied with location on the WEB to optimize stiffness and reduce mass. Interior and exterior close-outs and stiffeners as well as the cable tunnel are not shown in Figure 4. Nominal aerogel thickness for the side walls are 25 mm, and 32 mm for the bottom wall. The solar array structure and cable tunnel also not shown. The solar array consists of a honeycomb sandwich panel with a 32 mm thick glass-epoxy covered solid aerogel plug which fits into the top of the WEB. The honeycomb cells directly above the WEB are also filled with aerogel. The role of the cable tunnel seen at the front of the Rover in Figure 1 is to provide additional thermal insulation for the cabling that must pass from the interior electronic to exterior components. The copper conductors in the cabling are a significant heat loss path to the outside of the rover, Loss down the cables is decreased by a factor of 15 by providing insulation over a much longer length of the cable than would be obtained by a direct pass through.

The insulation in the cable tunnel consists of Eccofoam polyurethane foam. The cable tunnel provides a 0.5 meter insulated path.

The solid silica aerogel performs as the main thermal insulation, with the thermal design tailored to minimize heat loss through the vertical spars. The silica aerogel was manufactured at the Jet Propulsion Laboratory (JPL) at a density of 15-20 mg/cc (0.94 -1.25 lb/ft³). The aerogel was available at low densities because of a JPL development effort to use silica aerogel for cosmic dust capture experiments[5]. The lowest available density available from outside researchers was $\approx$ 50 mg/cc. Although the 15-20 mg/cc density is not the minimum for thermal conductivity as a function of density, it was chosen for the more important need to reduce overall system mass. There is extensive experience at JPL in manufacturing aerogel in this density range and this aerogel has been used on several Space Shuttle flight experiments. This silica aerogel is transparent because it was developed for particle dust experiments. To reduce internal radiation, a 5 mil gold coated kapton film was placed between the two layers of solid silica aerogel to act as a radiation barrier. The solid aerogel used in the Mars Rover is hydrophobic, and has good mechanical shock resistance. The hydrophobic quality of the silica aerogel is a result of the different processing chemistry used at JPL. Conventional chemical processes used to produce silica aerogel produce a material which is hydrophilic.

The open cavity sheet and spar design was dictated by the mechanical properties and handling ability of the aerogel. While the aerogel can mechanically withstand the loading from launch vibration and shock from the Mars surface landing, it is essentially a brittle-rubbery material at the low densities used in the Mars Rover. The material has good compressive properties, but low shear strength. This necessitated that the WEB be built around the aerogel in a sequential manner. Since the aerogel cannot handle large shear deformations without breaking, the sheet and spar WEB design allows the partially assembled structure to be deformed to allow the insertion of the aerogel. Once the aerogel is in place, the second facesheets and close-outs are bonded in place to rigidize the structure. Fully assembled, the WEB has sufficient strength and stiffness to meet a 75 G launch and impact design requirement.

THERMAL PERFORMANCE OF THE AEROGEL AND WEB DESIGN

The heat transfer through a specified volume is comprised of the summation of heat transfer by conduction, convection and radiation. This design was developed to minimize the total heat transfer. For many designs, convective heat transfer can be significant at pressures down to 1 torr, and the Mars surface

environment is nominally 8-10 torr CO₂. For this design, the convective heat transfer was eliminated entirely by filling the volume between the spars with solid aerogel. There is no internal convection within the aerogel because the interstitial spacing within the aerogel lattice structure is order's of magnitude less than the mean free path of a molecule at low pressures and vacuum. The largest constituent for heat transfer then becomes conduction. This was minimized by designing the structure with low conductivity glass-epoxy composite and minimizing the cross-sectional surface area for conduction through the spars. The facesheets are low conductivity glass/epoxy composites to minimize conduction through the spars. The remaining aspect of heat transfer is thermal radiation. Internal radiation was reduced by placing a layer of gold coated kapton between the two layers of aerogel. In the future, the aerogel may be directly opacified, eliminating the need for an additional radiation barrier between the sheets of the aerogel. External radiation was reduced by having the exterior glass epoxy facesheets covered with a low emissivity coating, in this case co-cured gold coated kapton. Interior surfaces had a high emissivity surface comprised of nickel coated graphite felt which was co-cured with the glass epoxy. This acts as a high emittance surface for the interior of the WEB, and also serves as EMI shielding and ESD protection.

The initial thermal property performance of this design was tested at the coupon level using ASTM C177-85, "Steady State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded Heat Plate Method[6]." This test evaluated a matched set of eight inch diameter, one inch thick coupons of the configuration shown in Figure 3. Figure 5 shows a comparison in the performance of the of the solid aerogel design including the structure (4.25 lb/ft³ density) with the previous opacified aerogel design (9.0 lb/ft³ density) at 10 torr CO₂ pressure. Thermal conductivity performance at 1 atmosphere and vacuum are presented in Table 1. As seen from Figure 5, the solid aerogel structure has a lower thermal conductivity than the previous state of the art coupled with a much lower density. Included for reference are the results for two nonstructural materials which are considered to be representative of state of the art foam and a fibrous insulation. The materials are Eccofoam SH, a low conductivity foam and TG-1500, a glass fiber insulation, each at 32 mg/cc (2.0 lb/ft³) densities. Both were tested under the same conditions. The use of these materials as part of a structural assembly would increase the integrated structure density to 120 - 150 mg/cc (7.5 -9.3 lb/ft³) because their higher conductivity would require more volume and higher density of insulation. The additional volume would take up critical space needed for electronics. Furthermore, additional insulation would have to be added to compensate for increased structural heat losses. Foam insulation was used selectively in nonstructural areas. The Eccofoam SH (2 lb/ft³ density)

was used in the cable tunnel and Rohacell 31 (3.1 lb/ft³ density) is being used to hold in place the RHU's within the axial glass-epoxy tube.

One key feature of solid aerogel as a thermal insulator is that the gas conduction portion of the heat transfer mechanism across the aerogel layer is greatly reduced even at a 10 torr gas pressure level. Figure 6 compares the apparent thermal conductivity for the two versions of WEB panel designs: an early design of Nomex honeycomb structure with Syloid 244 aerogel powder and aluminum powder opacification and the new sheet and spar design with solid aerogel. In addition to the weight advantage, WEB panels with solid aerogel also display far less difference in thermal conductance between the vacuum condition and the 10 torr condition, as compared its the counterpart using Syloid powder. Gas conduction in an insulating system is governed by the characteristic length of the solid and the gas mean freepath. For CO₂ at 10 torr the gas mean freepath is in the micron range. The microstructure of the solid aerogel appears to offer significantly better characteristics to suppress gas conduction than a powder insulation.

It should be noted that there is also a profound difference between the opacified powder system and the solid aerogel system with respect to the sensitivity to temperature increase. The solid aerogel panel is not opacified and shows a strong performance degradation at higher operating temperatures. Many different opacification schemes can be explored to affect the temperature dependence of the solid aerogel insulation. However, the design in its current form, appears to be more suitable for the Mars Rover's application. As shown in Figure 6, in comparison to the opacified powder aerogel system, the solid aerogel WEB design offers approximately 30% more thermal resistance at -40°C and 10 torr than the opacified powder aerogel system. On the other hand, at 25°C/vacuum, the solid aerogel WEB would only have half the thermal resistance (or double the effective conductance) of its aerogel powder counterpart. This feature allows the batteries to remain under 40 °C with the 3 watts RHU heating load in a 25 °C integration and ground pad environment without air conditioning.

Building upon the results from the coupon level testing, a full scale engineering model was fabricated and evaluated for both its thermal and mechanical performance. The Engineering Model (EM) tests consisted of a series of three steady state thermal balance tests (EM1, EM2, and EM3) which were successfully concluded in January 1995. The objective of the EM1 test sequence was to verify the WEB/aerogel design concept. The purpose of the EM2 test sequence was to demonstrate the viability of using a thermal link to control the WEB temperatures under cruise/vacuum conditions. The purpose of the EM3 test sequence was to assess the effects of vibration testing on aerogel thermal properties and performance. The

engineering model was tested in a thermal vacuum chamber that had a test section of approximately 1 meter in diameter by 1.5 meters in height. The test section temperature was maintained by separate temperature controlled top, middle, and door shrouds. Temperature capability ranged from -100° C to +40° C. The chamber had the capability of achieving a vacuum of 10⁻⁶ torr vacuum or of maintaining a low pressure (4 torr to 10 torr) CO₂ or GN₂ environment. This report describes the Mars Rover WEB Thermal Model and the actual thermal testing of the Mars Rover WEB. The motivation for each test, as well as a comparison of pre-test thermal model nodal temperature predictions with actual temperatures are presented.

1: EM1 TEST: WEB WITHOUT INTERIOR COMPONENTS

The objective of the Mars Rover WEB model test was to evaluate the thermal performance of the solid Aerogel WEB design under critical simulated mission environmental conditions. These conditions included a cruise/vacuum and a representative Mars surface condition with 8-torr CO₂ over temperature extremes. In order to evaluate the conductive characteristics of the aerogel, only the steady state test data were analyzed. The engineering model consisted of the WEB topped with a solar array substrate with a high emissivity coating. No interior components or mass simulators were included in the EM1 test. Within the WEB was placed an aluminum box containing a 5-watt heater to simulate the heat-generating capability of the RHU's plus the internal rover electronics. The engineering model was instrumented with 15 thermocouples. Gold and black reference plates with heaters were also included inside the test chamber.

The Rover WEB was analyzed with an 89 node SINDA model with 246 linear conductors and 122 radiative couplings. Relevant thermal parameters used in the analysis include the effective thermal conductivity of the aerogel with structure, the thermal conductivity of the crushed aerogel in the solar array honeycomb structure, the thermal conductivity of the glass-epoxy structure (WEB surfaces including edge effects), the surface emittances of the external surface (gold coated Kapton), the interior surfaces (Nickel coated graphite) and the free convention coefficient of CO₂ at 10 torr pressure. Table 2 presents these parameters used in the analysis to assess the upper and lower temperature limits.

The engineering model was suspended in the test chamber with the solar array surface facing the chamber door. The pre-planned pressure/ temperature/ power profiles were then imposed. The actual profiles consisted of a sequence of five sub-profiles, each of which consisted of a transient terminating with a set of steady state pressure/ temperature/ power values. The first test case simulates the situation where the Rover is

inside the Pathfinder-lander during cruise/vacuum and attached to a lander petal. The thermal strap is designed to conduct, under nominal near-Earth conditions, 2 watts from the interior to the 5°C lander fluid-loop cold-finger, leaving an effective 1 watt from the RHU. The second test case was a deviation from the original test plan. The test condition was at 10^{-2} torr for the first half of the test period instead of the intended 8 torr to 10 torr pressure level. This was not intended but was a consequence of CO₂ condensing on the cold LN₂ shroud inlet. The chamber was then allowed to stabilize before proceeding to the next test segment. The following cases simulate operations on the Mars surface, with the 0 °C to simulate the upper case temperature boundary for operation during the day, and -80 °C being the lower case boundary for operation during night. The power settings to the heater simulates the residual heat available for heating and also to provide a heating power dependency for the analytical model.

The test results are summarized in Figure 7, which compares the measured interior of the WEB, with steady state predictions. It can be seen that for the cruise/vacuum simulation, where overheating is a concern, the measured interior temperature is close to the lower bound of the prediction. On the other hand, for the Mars night simulation where cold protection is the emphasis, the measured interior temperature is close to the upper temperature prediction level. In other words, the test data demonstrate that the WEB thermal design using solid aerogel is sufficient for the cold protection on the Mars surface and yet would not be over insulated during cruise vacuum.

The EM1 thermal test was a successful demonstration of the thermal performance of the aerogel design within the limitations of the test set-up. One limitation was the inability to maintain a uniform temperature gradient across the shroud. The test chamber was not designed specifically for low pressure testing. There was also difficulty in maintaining the 10 torr CO₂ pressure at the low temperatures. This resulted in CO₂ solidifying at on the liquid nitrogen cooling lines (Dry ices forms at -120°C at 8 torr and -134°C at 1 torr). These experimental difficulties were resolved in the subsequent set of testing.

2: EM2 AND EM3 TESTS: WEB WITH INTERIOR COMPONENTS

The object of the EM2 test was to confirm EM1 test results using a more complete test article (i.e., WEB with interior components) and to validate the thermal design of the Rover interfacing components. The heater box of EM1 was replaced with Rover interior component mock-ups including a RHU heater simulator. Additionally, a realistic cable system with a cable tunnel box and internal and external thermal straps were also added.

In addition to modifying the Rover engineering model, several changes were made to the chamber environment, to the gold reference plate, and to the method of suspending the engineering model in the chamber. A special suspension fixture was fabricated to not only suspend the engineering model but to also to support the Rover/Lander thermal interface with a 10 pound preload. The interface was covered with CHO-THERM 1679 to enhance thermal conductance. Chamber cooling loops were modified to improve uniformity. The WEB engineering model and chamber were instrumented with a total of 34 thermocouples.

The EM2 testing was conducted in five sequential stages representing A: ground launch pad operations, B: cruise, C: first day on Mars on the Pathfinder Lander petal, D: free mobility on Mars and E: a repeat of the vacuum condition from the EM1 test. The RHU heater simulator supplied a continuous 2.76 watts during phases A through D, and one watt during phase E. Phases D and E were to be repeated after vibration tests in order to assess the effect, if any, of vibration on the thermal performance of the solid aerogel. The detailed results for the EM2 thermal vacuum testing will not be presented. The testing validated the performance of the aerogel design, and provided excellent correlation with the thermal model for the ground conditions and cruise condition performance but showed that there was a need for improvement for the thermal strap design for the interface between the Rover WEB and the Pathfinder petal.

The EM3 test was conducted after the Rover engineering model was subjected to Random and Sine vibration to simulate launch loads, and pulse tests to simulate Mars landing conditions. These tests were completed on the engineering model with full mass simulators for interior components and a mock-up of the tie-down attachment to the lander petal. These loads consisted of 30 G random and sine vibration in each of the three axis, and a 50 G shock load in each of the three axis. The EM3 tests were conducted similarly to the conditions to EM2. The thermal showed that there was no thermal performance degradation of the aerogel within measurement tolerances after being subjected to structural loads.

Based on the excellent performance of the Rover engineering model, minor modifications were made in the structural design of the WEB. For the thermal design, the basic concept was validated, and the design for the thermal link to lander was modified and improved. The Mars Rover qualification unit was completed in February 1995, and will undergo thermal and environmental testing during April and May 1995. The completion of the Mars Rover Flight Unit is scheduled for June 1995.

SUMMARY

A integrated lightweight structure and thermal insulation have been developed for the Mars Rover Warm Electronics Box. It combines a low thermal conductivity fiber reinforced composite structure with co-cured thermal control surfaces and an ultra-lightweight hydrophobic solid silica aerogel. This eliminates convection and minimizes conduction and radiation. It provides excellent thermal insulation both at low gas pressures and in vacuum; and meets the structural requirements for spacecraft launch loads and for a 50 g impact landing at Mars without damage to the insulation or structure. The integrated design provides for variable conductance to meet the different thermal requirements for the different phases of operation. The system incorporates heat conduction paths to provide thermal control on the ground and during cruise, which are then disconnected for Mars operation. The thermal performance of the design was evaluated at prototype level, and as a full scale structural engineering model. This new design is being flight tested during the next year and is being considered for other planetary missions.

REFERENCES

1. "Martian Soft Lander Insulation Study," O.J. Wilbers, B.J. Schelden, J.C. Conti, Heat Transfer and Thermal Control. Progress in Astronautics and Aeronautics, J.W. Lucas, Ed. Vol. 24, MIT Press, pp. 630-658,(1970).
2. Owens-Corning Publication No. 1-AUR-18927, 1994
3. "Aerogels," J.R. Fricke, Scientific American, Vol. 258, No. 5, pp. 92-97, May 1, 1988.
4. "Thermal Conductivity of Monolithic Organic Aerogels," J.R. Fricke, *et. al.*, Science, Vol. 255, pp. 971-972, February 21, 1992.
5. "P. Tsou, Jet Propulsion Laboratory, personal communication.
6. ASTM C177-85, "Steady State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded Heat Plate Method."

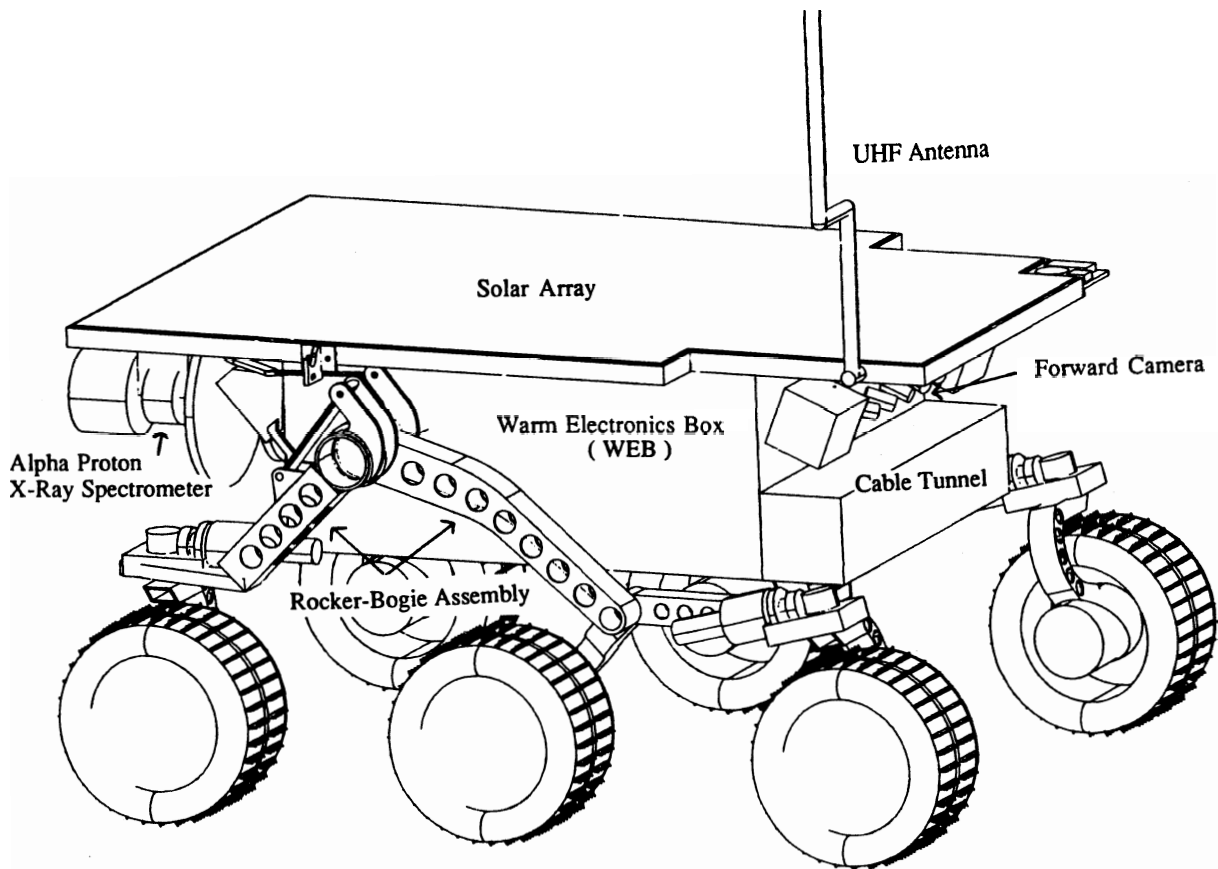


Figure 1: Schematic drawing of the Mars Rover.

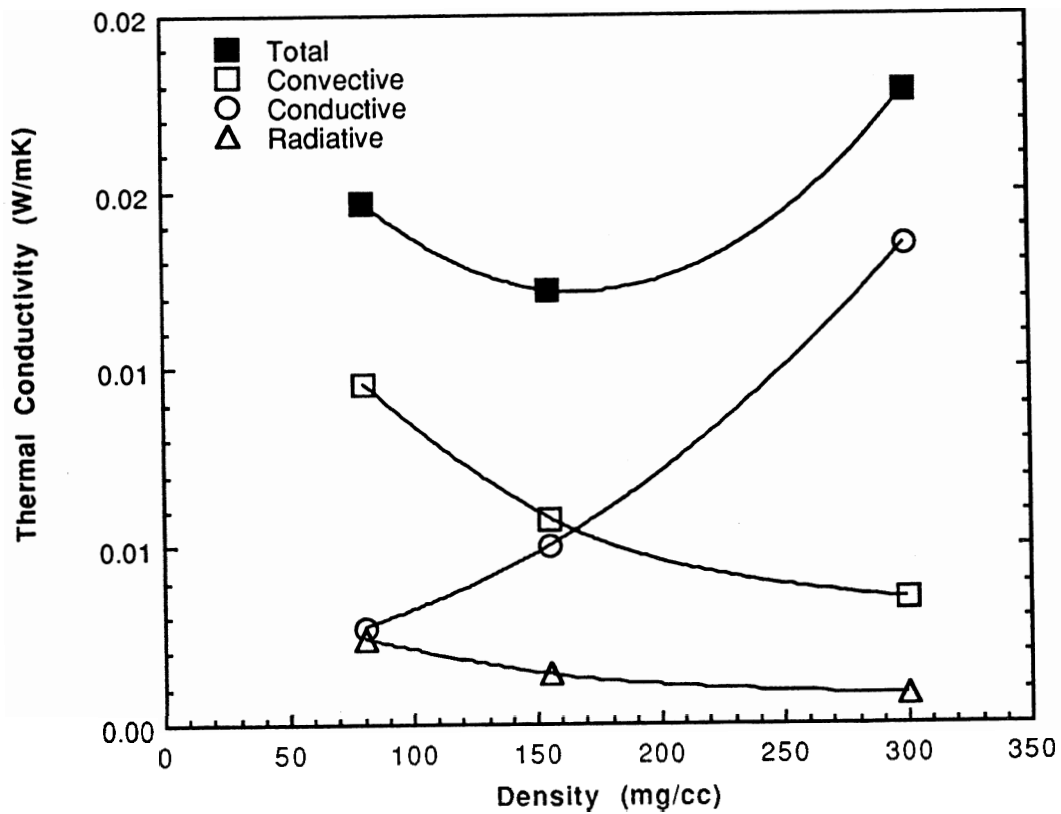


Figure 2: Thermal conductivity of silica aerogel.

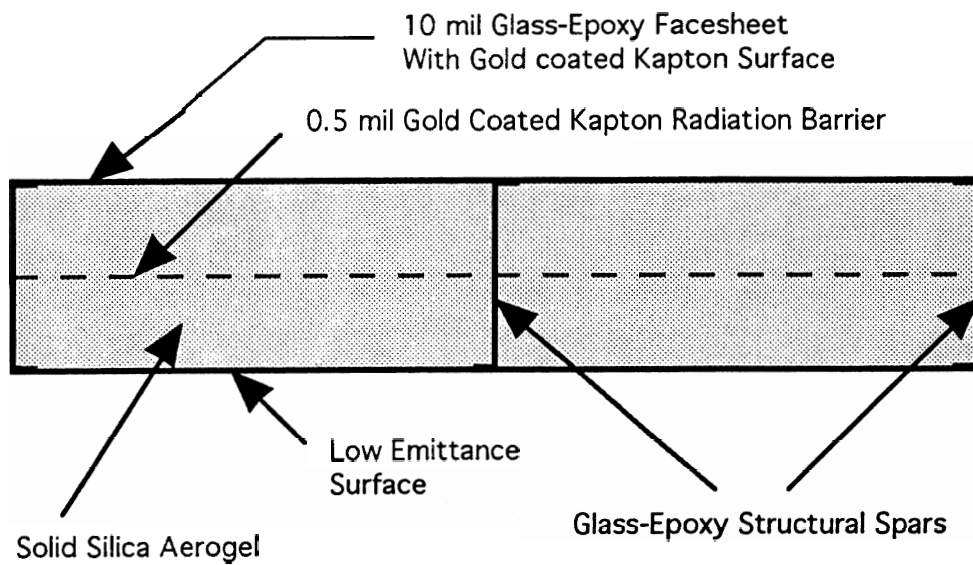


Figure 3: Configuration of aerogel structural insulation assembly.

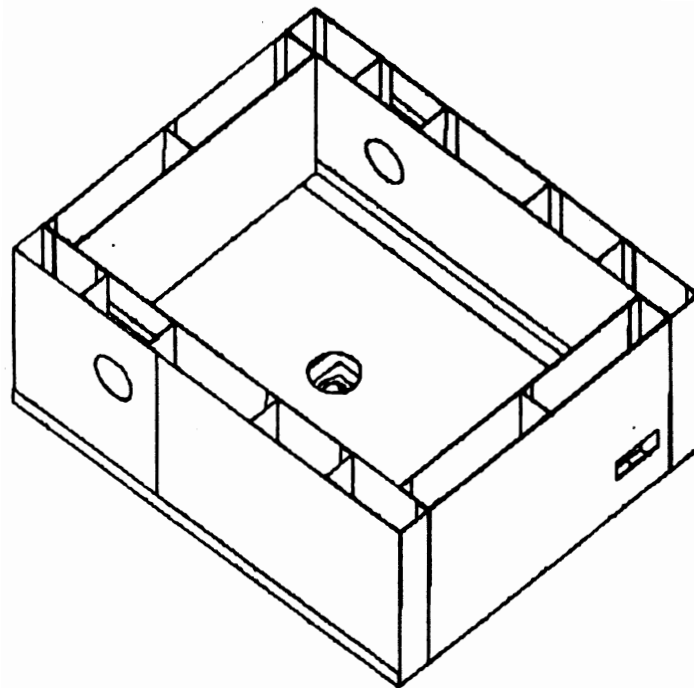


Figure 4: Isometric view of the Rover WEB sheet and spar design without close-outs and cable tunnel.

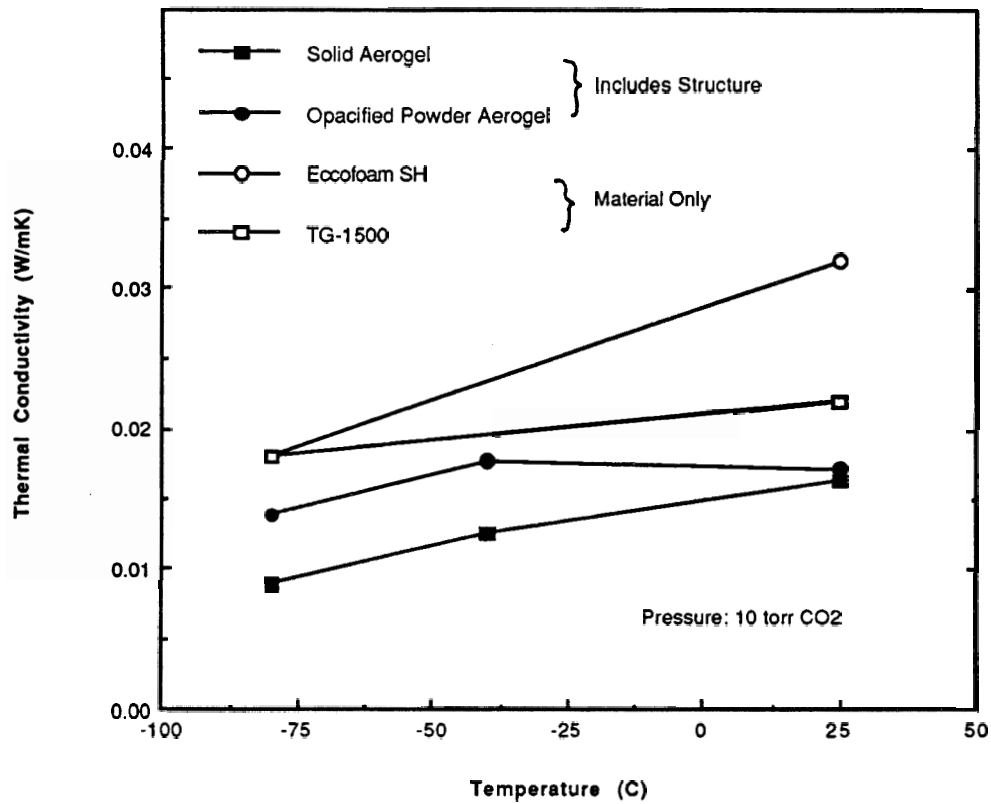


Figure 5: Thermal conductivity of aerogel structural insulation and comparative insulation materials.

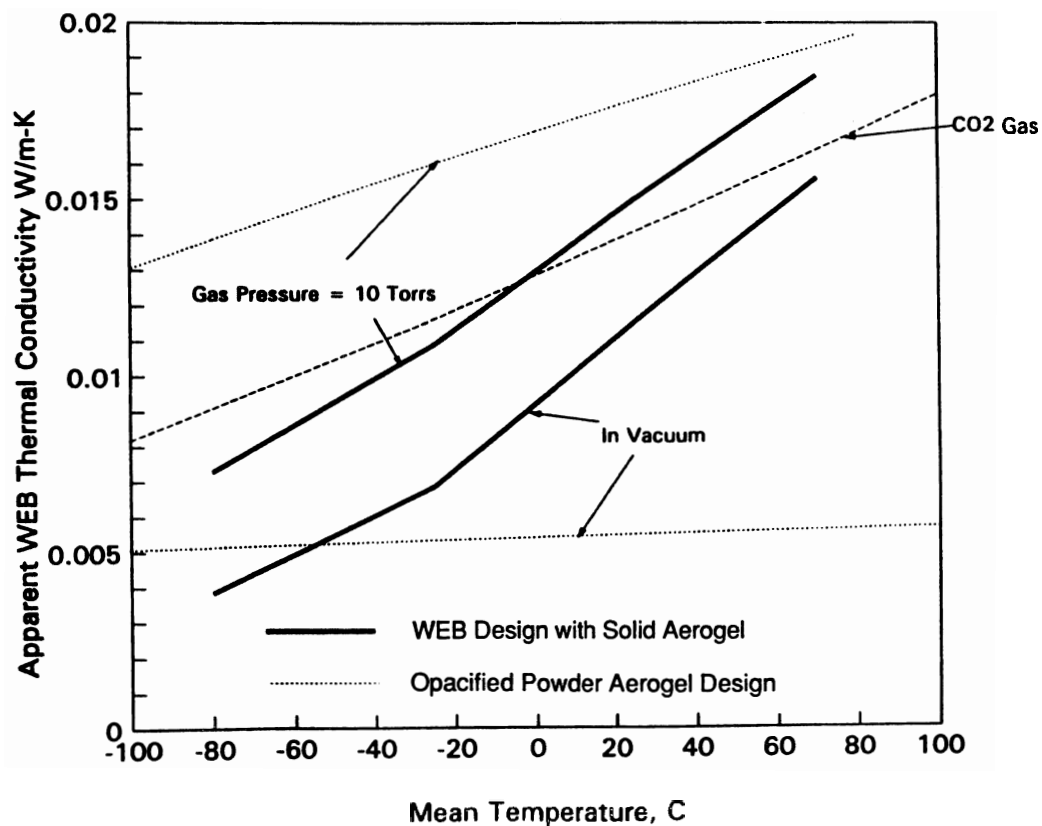


Figure 6: Comparison of Rover WEB performance with solid silica aerogel versus opacified powder design.

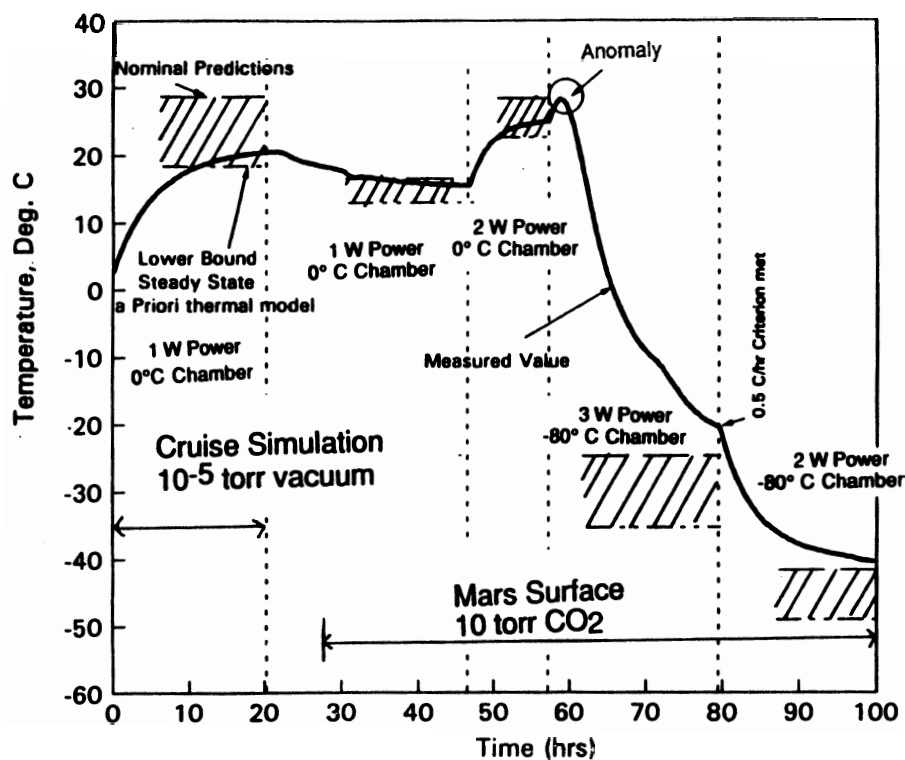


Figure 7: Comparison of Rover WEB interior temperature performance verses analytical model.

Table 1: Thermal Conductivity measurements of the Aerogel with Glass-epoxy structure at the coupon level.

Condition	Temperature (°C)	Thermal Conductivity (Watts/meter/Kelvin)
1 atm	25	0.0333
10 torr CO ₂	-79	0.0089
10 torr CO ₂	-27	0.0126
10 torr CO ₂	25	0.0163
1.2 x 10 ⁻³ torr Nitrogen	-78	0.0049
1.2 x 10 ⁻³ torr Nitrogen	-26	0.0074
0.4 x 10 ⁻³ torr Nitrogen	25	0.0113

Table 2: Physical parameters used in the SINDA model

Parameter	Upper Temperature Bound Case	Lower Temperature Bound Case
Effective Thermal Conductivity (W/mK)		
Crushed aerogel at 25°C :		
10 torr	0.0150	0.0190
Vacuum	0.0087	0.0105
Solid aerogel at 25°C :		
10 torr	0.0143	0.0170
Vacuum	0.0096	0.0115
Glass-Epoxy	0.28	0.70
Infrared Emittances		
Gold Coated Kapton	0.02	0.05
Nickel Coated Graphite	0.75	0.84
Free Convection of CO ₂ at 10 torr (W/m ² -K)	0.2	0.6